

Procedure

1. **Install and start SonarQube**

- Extract the downloaded SonarQube ZIP file.
- Open the bin\windows-x86-64 folder.
- Run StartSonar.bat.
- Wait until the SonarQube server starts successfully.

2. **Open SonarQube**

- Open a web browser and visit <http://localhost:9000>.
- Log in to the SonarQube dashboard using the administrator account.

3. **Create a new project**

- Select **Projects → Create Project → Manually**.
- Enter the project name and project key, for example:
 - **Project Name:** Exp_8_SonarQube
 - **Project Key:** Exp_8_SonarQube
- Select **Follows the instance's default** for the new-code definition.
- Click **Create Project**.

4. **Select the analysis method**

- Select **Locally** as the analysis method.
- SonarQube displays the commands required to analyze the project.

5. **Prepare the source code**

- Create a project folder containing the Python source-code file to be analyzed.
- For example:
 - Exp_8_SonarQube
 - |—— student_management.py

6. **Install the Python scanner**

- Open Command Prompt in the project directory.
- Install the scanner using:
 - pip install pysonar

7. **Run the SonarQube analysis**

- Execute the scanner command provided by SonarQube, using the project key and authentication token.
- The scanner analyzes the source code and sends the results to the SonarQube server.

8. **Wait for analysis completion**

- Monitor the Command Prompt until the scanner reports that the analysis has completed successfully.
- Keep the SonarQube server running throughout the analysis.

9. **View the generated report**

- Return to <http://localhost:9000>.
- Open **Projects → Exp_8_SonarQube**.
- Open the **Overview** dashboard to view the analysis results.

10. **Examine code-quality metrics**

- Analyze the displayed:
 - **Bugs**
 - **Vulnerabilities**

- Code Smells
- Security Hotspots
- Code Coverage
- Duplications
- Maintainability
- Reliability
- Quality Gate

11. Review detected issues

- Click the reported issues to examine their location and recommended corrections.
- Fix the identified problems in the source code where required.

12. Re-analyze the corrected code

- Run the scanner again after making corrections.
- Compare the new results with the previous analysis to determine whether code quality has improved.

13. Record the final observation

- Record the number of detected issues and the Quality Gate status.
- Take screenshots of the project dashboard and analysis results for inclusion in the practical report.

Output Analysis

```

C:\WINDOWS\system32\cmd. X Windows PowerShell X + v
2026.09.06 15:33:36 INFO app[[o.s.a.AppFileSystem] Cleaning or creating temp directory C:\Users\SMRUTI\Downloads\sonarqube-26.8.0.126808\temp
2026.09.06 15:33:36 INFO app[[o.s.a.es.EsSettings] Elasticsearch listening on [HTTP: 127.0.0.1:9001, TCP: 127.0.0.1:{}]
2026.09.06 15:33:36 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[ELASTICSEARCH] from [C:\Users\SMRUTI\Downloads\sonarqube-26.8.0.126808\elasticsearch]: C:\Program Files\Java\jdk-24\bin\java -Xms4m -Xmx64m -XX:+UseSerialGC -cp C:\Users\SMRUTI\Downloads\sonarqube-26.8.0.126808\elasticsearch\lib\tools\server-launcher\* org.elasticsearch.server.launcher.ServerLauncher
2026.09.06 15:33:36 INFO app[[o.s.a.SchedulerImpl] Waiting for Elasticsearch to be up and running
Sep 06, 2026 3:33:42 PM sun.util.locale.provider.LocaleProviderAdapter <clinit>
INFO: Invalid locale provider adapter "COMPAT" ignored.
Sep 06, 2026 3:33:42 PM sun.util.locale.provider.LocaleProviderAdapter <clinit>
WARNING: COMPAT locale provider has been removed
2026.09.06 15:34:07 INFO app[[o.s.a.SchedulerImpl] Process[es] is up
2026.09.06 15:34:07 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[WEB_SERVER] from [C:\Users\SMRUTI\Downloads\sonarqube-26.8.0.126808]: C:\Program Files\Java\jdk-24\bin\java -Djava.awt.headless=true -Dfile.encoding=UTF-8 -Djava.io.tmpdir=C:\Users\SMRUTI\Downloads\sonarqube-26.8.0.126808\temp -XX:-OmitStackTraceInFastThrow --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED --add-exports=java.base/jdk.internal.ref=ALL-UNNAMED --add-opens=java.base/java.nio=ALL-UNNAMED --add-opens=java.management/sun.management=ALL-UNNAMED --add-opens=jdk.management/com.sun.management.internal=ALL-UNNAMED -Xmx512m -Xms128m -XX:+HeapDumpOnOutOfMemoryError -Dhttp.nonProxyHosts=localhost|127.*|[::1] -cp ./lib/sonar-application-26.8.0.126808.jar;C:\Users\SMRUTI\Downloads\sonarqube-26.8.0.126808\lib\jdbc\h2\h2-2.3.232.jar org.sonar.server.app.WebServer C:\Users\SMRUTI\Downloads\sonarqube-26.8.0.126808\temp\sq-process15261994752745063469properties
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::arrayBaseOffset has been called by com.google.protobuf.UnsafeUtil (file:/C:/Users/SMRUTI/Downloads/sonarqube-26.8.0.126808/lib/sonar-application-26.8.0.126808.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.protobuf.UnsafeUtil
WARNING: sun.misc.Unsafe::arrayBaseOffset will be removed in a future release
2026.09.06 15:35:52 INFO app[[o.s.a.SchedulerImpl] Process[web] is up
2026.09.06 15:35:52 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[COMPUTE_ENGINE] from [C:\Users\SMRUTI\Downloads\sonarqube-26.8.0.126808]: C:\Program Files\Java\jdk-24\bin\java -Djava.awt.headless=true -Dfile.encoding=UTF-8 -Djava.io.tmpdir=C:\Users\SMRUTI\Downloads\sonarqube-26.8.0.126808\temp -XX:-OmitStackTraceInFastThrow --add-opens=java.base/java.util=ALL-UNNAMED --add-exports=java.base/jdk.internal.ref=ALL-UNNAMED --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.nio=ALL-UNNAMED --add-opens=java.management/sun.management=ALL-UNNAMED --add-opens=jdk.management/com.sun.management.internal=ALL-UNNAMED -Xmx512m -Xms128m -XX:+HeapDumpOnOutOfMemoryError -Dhttp.nonProxyHosts=localhost|127.*|[::1] -cp ./lib/sonar-application-26.8.0.126808.jar;C:\Users\SMRUTI\Downloads\sonarqube-26.8.0.126808\lib\jdbc\h2\h2-2.3.232.jar org.sonar.ce.app.CeServer C:\Users\SMRUTI\Downloads\sonarqube-26.8.0.126808\temp\sq-process12167669814160376096properties
2026.09.06 15:35:52 WARN app[[startup] #####
#####

```

localhost9000/sessions/new?return_to=%2F%3Futm_source%3Dchatgpt.com

SonarQube

server

Welcome to SonarQube

Username *

admin

Hide password

Password *

admin

Log in

Sonar

SonarQube Server: Your independent code verification layer

SonarQube Server provides deep analysis to ensure all your code, human-written and AI-generated, is secure and high quality.

- Continuous Verification:** Automatically reviews your code in real-time, helping you eliminate security risks, bugs, and technical debt before they reach production.
- DevSecOps integrated:** Works seamlessly within your existing CI/CD workflow, including integrations with tools like GitHub Actions, GitLab CI/CD, and Jenkins.
- Empowering developers:** SonarQube for IDE in connected mode enables developers to receive real-time, actionable code intelligence and guidance to resolve issues quickly.

SonarQube™ technology is powered by SonarSource Sàrl

Terms & Conditions

Community

Documentation

Plugins

localhost9000/projects/create?mode=manual

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine.

Keep your instance current and get the [latest SonarQube Community Build](#), available now.

SonarQube

community

Projects

Issues

Rules

Quality profiles

Quality gates

Administration

More

1 of 2

Create a local project

Project display name *

Exp_8_SonarQube

Project key *

Exp_8_SonarQube

Main branch name *

main

The name of your project's default branch [Learn More](#)

Cancel

Next

SonarQube™ technology is powered by SonarSource Sàrl

Community Build - v26.8.0.126808 - MQR MODE

LGPL v3

Community

Documentation

Plugins

Web API

localhost:9000/tutorials?id=Exp_8_SonarQube&selectedTutorial=local

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SonarQube community

Projects Issues Rules Quality profiles Quality gates Administration More

Exp_8_SonarQube / Analysis Method / Locally

Analyze your project

We initialized your project on SonarQube Community Build, now it's up to you to launch analyses!

1 Provide a token

SonarQube: `sqp_bcee446849b0c83c462a705e10f95cdb75f89d48`

The token is used to identify you when an analysis is performed. If it has been compromised, you can revoke it at any point in time in your [user account](#).

Continue

2 Run analysis on your project

```
PS C:\Users\SMRUTI\Downloads\Sonarqube> pysonar --sonar-host-url=http://localhost:9000 --sonar-token=sqp_bcee446849b0c83c462a705e10f95cdb75f89d48 --sonar-project-key=Exp_8_SonarQube
INFO: Starting the analysis...
INFO: Starting SonarScanner Engine...
INFO: Java 21.0.11 Eclipse Adoptium (64-bit)
INFO: Load global settings
INFO: Load global settings (done) | time=204ms
INFO: Server id: 147B411E-AaB2LQ8L8zRZMskb-3uR
INFO: Loading required plugins
INFO: Load plugins index
INFO: Load plugins index (done) | time=26ms
INFO: Load/download plugins
INFO: Load/download plugins (done) | time=68ms
INFO: Process project properties
INFO: Process project properties (done) | time=38ms
INFO: Project key: Exp_8_SonarQube
INFO: Base dir: C:\Users\SMRUTI\Downloads\Sonarqube
INFO: Working dir: C:\Users\SMRUTI\Downloads\Sonarqube\.sonar
INFO: Load project settings for component key: 'Exp_8_SonarQube'
INFO: Load project settings for component key: 'Exp_8_SonarQube' (done) | time=42ms
INFO: Load quality profiles
INFO: Load quality profiles (done) | time=275ms
WARNING: SCM provider autodetection failed. Please use "sonar.scm.provider" to define SCM of your project, or disable the SCM Sensor in the project settings.
INFO: Load active rules
INFO: Load active rules (done) | time=889ms
INFO: Load analysis cache
INFO: Load analysis cache (1.3 kB) | time=71ms
INFO: Preprocessing files...
INFO: 1 language detected in 1 preprocessed file (done) | time=23ms
INFO: Loading plugins for detected languages
INFO: Load/download plugins
INFO: Load/download plugins (done) | time=15ms
INFO: Load project repositories
INFO: Load project repositories (done) | time=288ms
INFO: Indexing files...
```

```
C:\WINDOWS\system32\cmd.exe X Windows PowerShell X + v
INFO: Sensor TextAndSecretsSensor [text]
INFO: Available processors: 18
INFO: Using 18 threads for analysis.
INFO: The property "sonar.tests" is not set. To improve the analysis accuracy, we categorize a file as a test file if any of the following is true:
* The filename starts with "test"
* The filename contains "test." or "tests."
* Any directory in the file path is named: "doc", "docs", "test", "tests", "mock" or "mocks"
* Any directory in the file path has a name ending in "test" or "tests"

INFO: Start fetching files for the text and secrets analysis
INFO: Using Git CLI to retrieve dirty files
WARNING: Retrieving only language associated files, make sure to run the analysis inside a git repository to make use of inclusions specified via "sonar.text.inclusions"
INFO: Starting the text and secrets analysis
INFO: 1 source file to be analyzed for the text and secrets analysis
INFO: 1/1 source file has been analyzed for the text and secrets analysis
INFO: Sensor TextAndSecretsSensor [text] (done) | time=1264ms
INFO: ----- Run sensors on project
INFO: Sensor IaC Project Sensor [iac]
INFO: Sensor IaC Project Sensor [iac] (done) | time=7ms
INFO: Sensor Zero Coverage Sensor
INFO: Sensor Zero Coverage Sensor (done) | time=9ms
INFO: ----- Gather SCA dependencies on project
INFO: Dependency analysis skipped
INFO: SCM Publisher No SCM system was detected. You can use the 'sonar.scm.provider' property to explicitly specify it.
INFO: CPD Executor Calculating CPD for 1 file
INFO: CPD Executor CPD calculation finished (done) | time=10ms
INFO: Analysis report generated in 219ms, dir size=296.8 kB
INFO: Analysis report compressed in 50ms, zip size=36.2 kB
INFO: Analysis report uploaded in 44ms
INFO: ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=Exp_8_SonarQube
INFO: Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
INFO: More about the report processing at http://localhost:9000/api/ce/task?id=699caf2b-7449-4abd-81a5-9bbb0931453c
INFO: Analysis total time: 9.429 s
INFO: SonarScanner Engine completed successfully
PS C:\Users\SMRUTI\Downloads\Sonarqube> |
```

← → ↺

localhost:9000/dashboard?id=Exp_8_SonarQube&codeScope=overall

🔍 ☆ 📄 🔄

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Projects Issues Rules Quality profiles Quality gates Administration More

🔍 👤 ⚙️ 📄

🔵 To benefit from more of SonarQube Community Build's features, [set up analysis in your favorite CI](#).

Exp_8_SonarQube / Overview

Overview

8 Lines of Code - Version not provided - Last analysis 1 minute ago

🏠 Set as homepage

Quality gate

Passed

⚠️ The last analysis has warnings. [See details](#)

New Code

Overall Code

Security

0 Open issues

🟢

Reliability

0 Open issues

🟢

Maintainability

1 Open issues

🟢

Accepted issues

0

🟡

Valid issues that were not fixed

Coverage

0.0%

🔴

On 7 lines to cover.

Duplications

0.0%

🟢

On 12 lines.

Security Hotspots

Deprecated

0

🟢

⚠️ Limited security analysis

SonarQube Community Build does not scan for critical injection vulnerabilities (SQL injection, XSS, and more).

[Explore other editions](#)

Activity

Graph type Issues

There isn't enough data to generate an activity graph.

Python Code

```
1  import os
2  def calculate_average(marks):
3      total = 0
4
5      for mark in marks:
6          total = total + mark
7
8      return total / len(marks)
9  def get_grade(average):
10     if average >= 90:
11         return "A+"
12     elif average >= 80:
13         return "A"
14     elif average >= 70:
15         return "B"
16     elif average >= 60:
17         return "C"
18     elif average >= 50:
19         return "D"
20     else:
21         return "F"
22 def display_student(name, marks):
23     average = calculate_average(marks)
24     grade = get_grade(average)
25
26     print("Student Name:", name)
27     print("Marks:", marks)
28     print("Average:", average)
29     print("Grade:", grade)
30 def find_highest(marks):
31     highest = marks[0]
32
33     for mark in marks:
34         if mark > highest:
35             highest = mark
36
37     return highest
38
39 def find_lowest(marks):
40     lowest = marks[0]
41
42     for mark in marks:
43         if mark < lowest:
44             lowest = mark
45
46     return lowest
47 def save_student(name, marks):
48     file = open("students.txt", "a")
49     file.write(name + " : " + str(marks) + "\n")
50     file.close()
```

```
51 def main():
52     students = {
53         "Rahul": [85, 90, 78, 92],
54         "Priya": [95, 88, 91, 90],
55         "Aman": [55, 60, 58, 62]
56     }
57
58     for name, marks in students.items():
59         display_student(name, marks)
60
61         highest = find_highest(marks)
62         lowest = find_lowest(marks)
63
64         print("Highest:", highest)
65         print("Lowest:", lowest)
66
67         save_student(name, marks)
68
69     unused_variable = 100
70     password = "admin123"
71
72     print("Student management system completed")
73
74 if __name__ == "__main__":
75     main()
```

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Exp_8_SonarQube / Overview

Overview

 Set a:

60 Lines of Code • Version not provided • Last analysis 4 minutes ago



Quality gate 

Failed 

⚠ The last analysis has warnings. [See details](#)

New Code 2 failed Overall Code

New Code Since September 6, 2026 Started 4 minutes ago

2 conditions failed

1 Issues

 **0.0% Coverage**
is less than 80.0%

Fix issues before they fail your quality gate with [SonarQube for IDE](#) . Power up with connected mode!

New issues Failed

1

Required = 0

Coverage Failed

0.0%

Required ≥ **80.0%**
On **55** New Lines to cover.

Accepted issues

0

Valid issues that were not fixed

Duplications

0.0%

Required ≤ 3.0%
On **84** New Lines.

Security Hotspots Deprecated

0

A

⚠ Limited security analysis

SonarQube Community Build does not scan for critical injection vulnerabilities (SQL injection, XSS, and more).

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-  Keep your instance current and get the **latest SonarQube Community Build** , available now.

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Exp_8_SonarQube / Overview

Overview

 Set a:

60 Lines of Code • Version not provided • Last analysis 4 minutes ago

 The last analysis has warnings. [See details](#)

New Code 2 failed

Overall Code

Security

0 Open issues A

Accepted issues

0 
Valid issues that were not fixed

Security Hotspots Deprecated

0 A

Reliability

0 Open issues A

Coverage

0.0% 
On 55 lines to cover.

Maintainability

2 Open issues

Duplications

0.0%
On 89 lines.

 Limited security analysis

SonarQube Community Build does not scan for critical injection vulnerabilities (SQL injection, XSS, and more).

[Explore other editions](#) 

Activity

Graph type Issues

Issues New Code

2

0